CLINICAL BINARY CLASSIFICATION :

Numerical Metrics

Split	ROC-AUC	Accuracy	Balanced Accuracy	Sensitivity / Recall	Specificity	Precision	F1-score	Brier Score	PR-AUC	TN	FP	FN	TP	Expected Calibration Error / ECE
validation	0.6522	0.8961	0.5000	1.0000	0.0000	0.8961	0.9452	0.0901	0.9029	0	8	0	69	0.0331
test	0.8438	0.8846	0.5000	1.0000	0.0000	0.8846	0.9388	0.0954	0.9733	0	9	0	69	0.0379

-ROC-AUC is below 0.5 on both validation and test, which is worse than random ranking.

- Balanced accuracy is exactly 0.5, meaning the average of sensitivity and specificity is no better than chance.

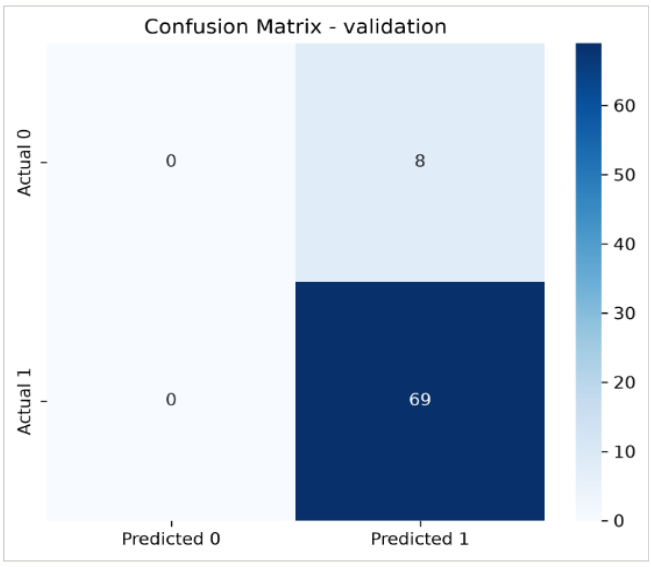
- Sensitivity is 1.0, but specificity is 0.0. The model is one-sided.

- PR-AUC is high only because the positive class is very common. The baseline positive rate is already about 0.89, so PR-AUC near 0.89 is not impressive.

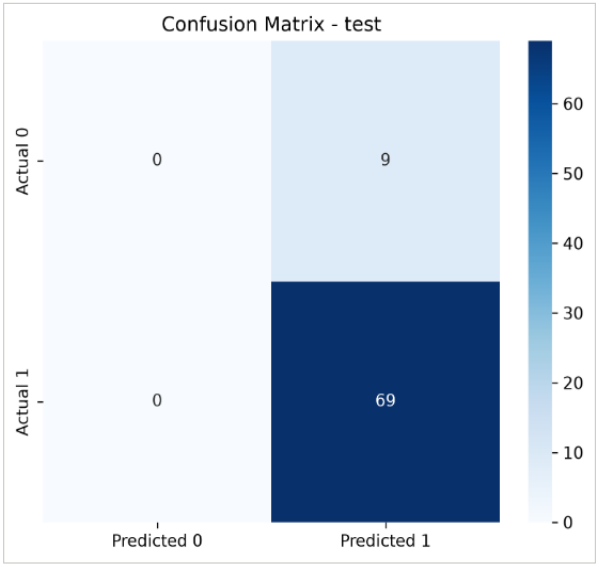
- Brier score and ECE are not terrible, but they are not enough to rescue the model because class discrimination is poor.

-Accuracy and F1 look strong, but they are misleading here because the data is heavily imbalanced toward HPV-positive patients. In validation, 69 of 77 patients are HPV-positive. In test, 69 of 78 patients are HPV-positive. A model that predicts almost everyone as HPV-positive can therefore obtain high accuracy, high precision, and high F1 without learning to recognize HPV-negative patients.

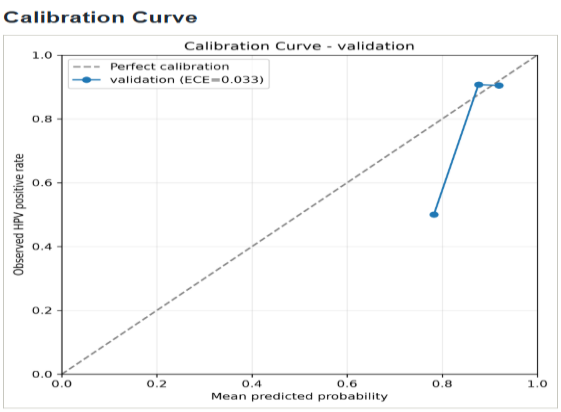
Confusion Matrix Heatmap

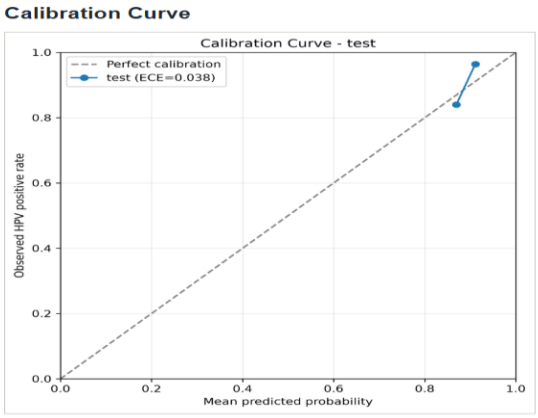


Confusion Matrix Heatmap

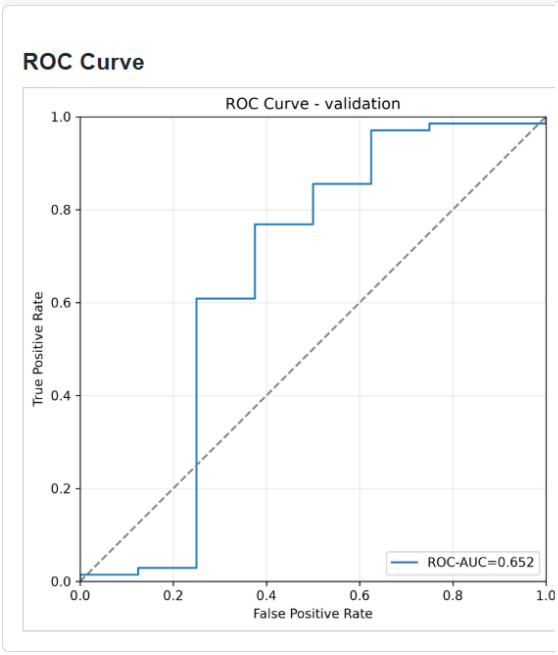


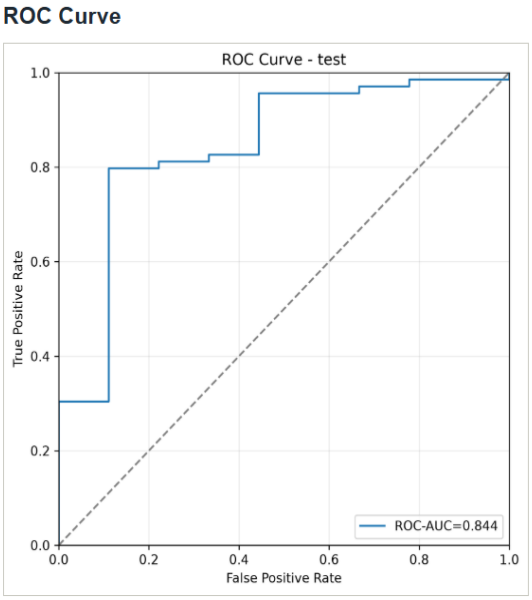
This is the clearest evidence that the clinical-only MLP is biased toward predicting HPV-positive. It catches all HPV-positive patients, but it completely fails on the HPV-negative class.





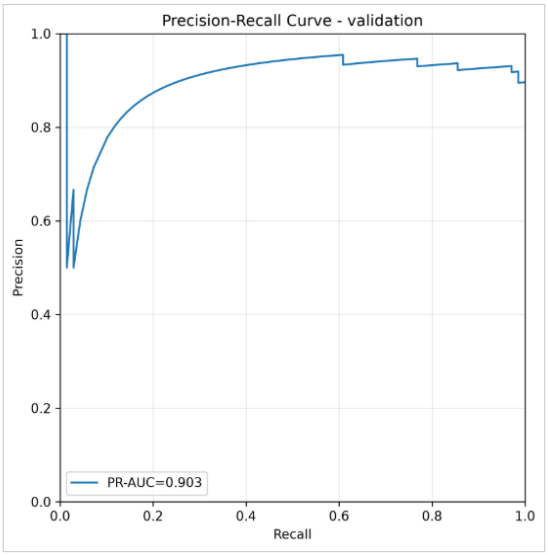
An ideal calibration curve follows the diagonal line, meaning predicted probability matches the observed HPV-positive rate. The clinical model's calibration curve is built mostly from high-probability bins. This means the model usually predicts high HPV-positive probability for everyone. In validation, the [0.9, 1.0] bin has mean probability 0.934 but observed positive rate 0.860, so the model is overconfident in that bin. In test, the [0.9, 1.0] bin has mean probability 0.933 but observed positive rate 0.875, again showing overconfidence.



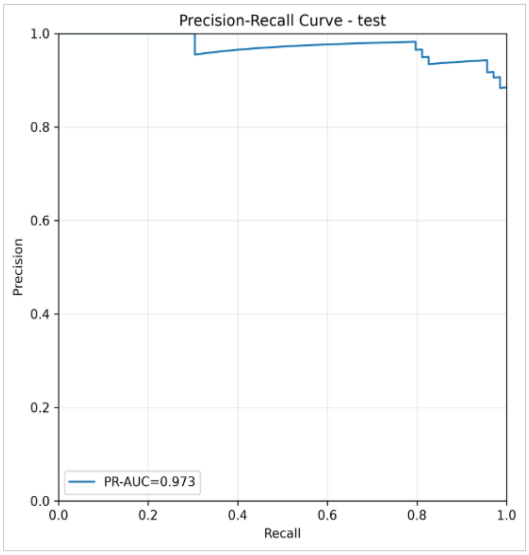


The ROC curve should ideally bend strongly toward the top-left corner. That would mean high true-positive rate and low false-positive rate across thresholds. In the clinical model, ROC-AUC is 0.429 on validation and 0.432 on test, so the ROC curve is weak and close to or worse than the random diagonal. This means the predicted probabilities do not rank HPV-positive and HPV-negative patients well.

Precision-Recall Curve

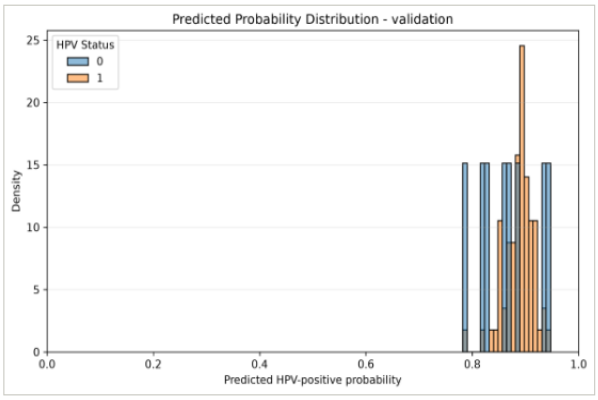


Precision-Recall Curve

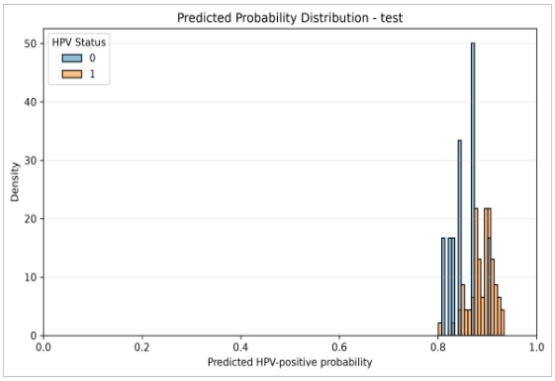


The PR curve may look acceptable because precision stays high, but this is mostly caused by the high number of HPV-positive cases. Since about 89 percent of the patients are HPV-positive, the baseline PR-AUC is already high. Therefore, PR-AUC values around 0.89 do not prove the model is strong. The PR curve should be interpreted together with specificity and balanced accuracy, which show the model is poor.

Probability Distribution

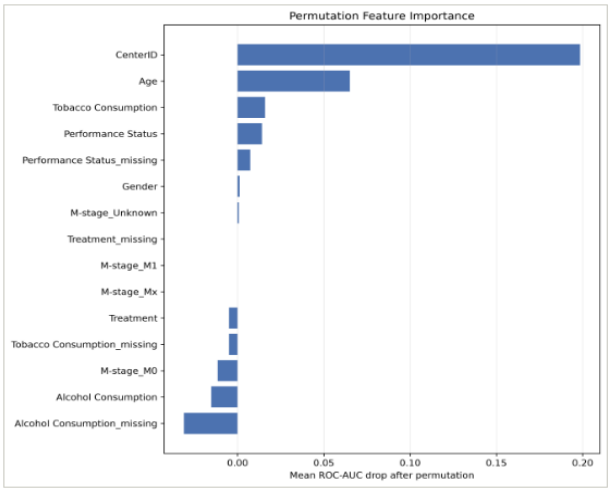


Probability Distribution

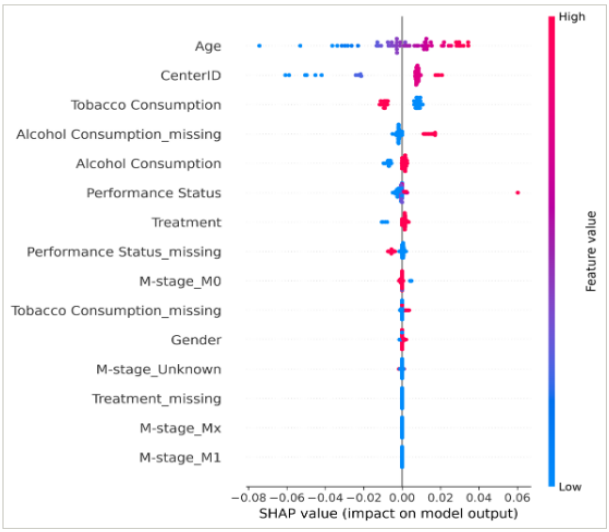


The ideal probability distribution plot would show HPV-negative patients concentrated near 0 and HPV-positive patients concentrated near 1. For the clinical model, both groups receive high predicted probabilities, so the distributions overlap heavily on the HPV-positive side. Because most predictions are above 0.5, the model classifies everyone as HPV-positive.

Feature Importance Plot



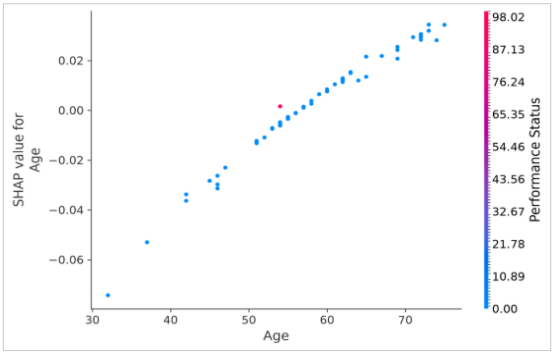
SHAP Summary Plot



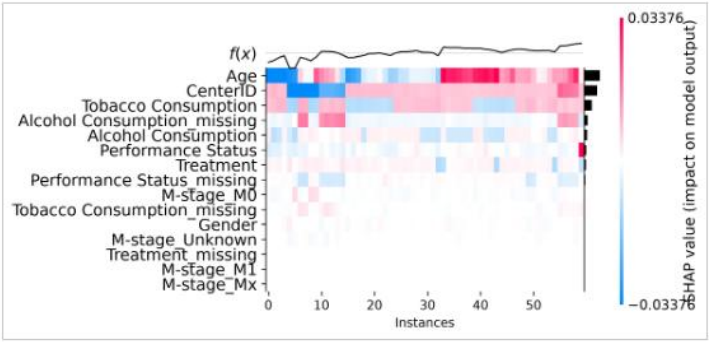
The permutation importance plot shows Age and Performance Status as the most influential clinical variables. However, the importance drops are small, meaning no clinical feature strongly controls classification performance. This supports the conclusion that the clinical-only feature set has limited discriminatory signal.

The SHAP summary plot also emphasizes Age, with smaller contributions from CenterID, Performance Status, Gender, Tobacco Consumption, and Treatment. The SHAP values are not large enough to create clear separation between classes. This means the model uses clinical variables, but their effects are weak.

SHAP Dependence Plot



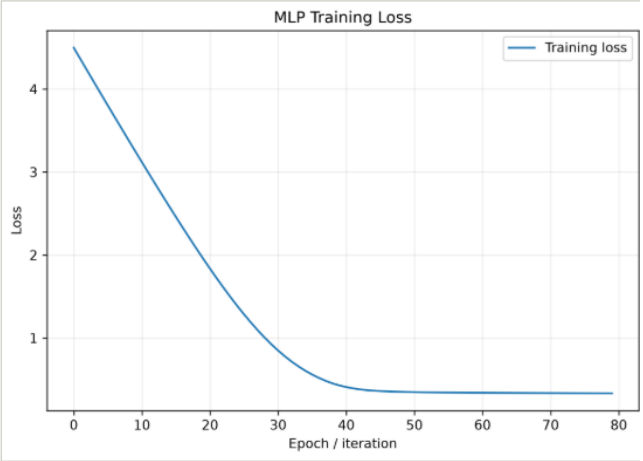
SHAP Heatmap Plot



The dependence plot automatically selected Age. This plot is useful for seeing how one feature changes the model prediction. In this case, Age has the strongest effect, but the overall model still fails to detect HPV-negative patients, so Age alone is not sufficient.

An ideal SHAP heatmap would show clear, consistent feature-effect patterns separating groups of patients. The clinical SHAP heatmap does not provide evidence of a strong separation pattern. The model's explanations are mostly small and concentrated around a few clinical variables.

Training Loss Curve



The training loss curve shows whether the MLP was able to optimize during training. Even if the training loss decreases, that does not mean the classifier is clinically useful. Here, the evaluation plots show that the model learned the majority-class pattern rather than a balanced HPV classifier.

Important note about functions or techniques used to help the model:

The MLP necessarily used basic neural-network components so it could train: ReLU activation in the hidden layers, sigmoid activation in the final binary output layer, and binary cross-entropy loss. These are standard minimum requirements for an MLP binary classifier, not special performance-boosting tricks.

We did not use extra functions or methods to help the model overcome the difficult dataset. For example, we did not use custom loss functions, focal loss, class weights, oversampling, undersampling, threshold tuning, hyperparameter tuning, dropout, batch normalization, L1/L2 regularization, learning-rate schedules, feature scaling, feature selection, or feature engineering. Because of this, the model was intentionally kept simple, but it also had no special support for the severe class imbalance or the raw feature-value scale. This is one reason the clinical model collapsed toward predicting the majority HPV-positive class.